NTNU
Norwegian University of Science and Technology
Faculty of Information Technology and Electrical Engineering
Department of Engineering Cybernetics

TERM PROJECT ASSIGNMENT

Name of candidate:	Anders Haarberg Eriksen
Topic:	Cybernetics and Robotics
Title of assignment (Norwegian):	Modellering og adaptiv regulering av et senedrevet hånd-eksoskjelett
Title of assignment (English):	Modelling and Adaptive Control of a Tendon-Driven Hand Exoskeleton

Assignment text:

Modern hand exoskeletons require precise and reliable control of the relevant mechanical states to enable natural, predictable, and safe interaction with the environment. The current implementation of the present glove–cable system introduces nonlinearities, uncertain mechanical parameters, and inconsistent finger positioning, making accurate control difficult. To address these challenges, an adaptive control framework is desired that is robust in the presence of parameter uncertainties and nonlinearities and ensures stable behavior during both free motion and object interaction.
The goal of this project is thus to design an adaptive controller for a 6-degree-of-freedom hand exoskeleton, capable of regulating both torque and fingertip positioning. This includes identifying relevant mechanical parameters and developing a nonlinear model of the system. Additionally, modifications to the glove mechanics and integration of fingertip sensors could enable better positioning and torque estimation and facilitate switching between control modes and should therefore be investigated. The final system should allow control of finger motion while providing insight into system properties such as friction and internal forces.
The work should thus include the following:

- Development of a control oriented, nonlinear model of a single finger.
- Design of an adaptive control architecture for finger position and/or contact force.
- Estimation of mechanical parameters and internal forces.
- Evaluation of fingertip sensors to support mode switching and refined control.
- Assessment of the results.

Assignment given:	12. January, 2026
Submission deadline:	1. June, 2026

Conducted at the Department of Engineering Cybernetics in collaboration with the N-Squared Lab in Erlangen, Germany

Supervisor(s):	Øyvind Stavdahl, NTNU Devon Rohlf, Friedrich-Alexander-Universität Erlangen-Nürnberg

Trondheim, 01.01.2026

Øyvind Stavdahl

Supervisor